Christmas lunch!

HE strategic session 3

11 December 2025

Agenda

Time	Topic
12:00 – 13:30	Christmas Lunch
13:30 – 14:00	Reflecting on Session 2:
14:00 – 14:40	Activity 1: HE research/education lines and topics
14: 40 – 15:00	*Break*
15:00 – 16:30	Activity 2: Designing the HE department of the future
16:30 – 16:45	Next steps
16:45	End of session

Summary session 1+2

Hydraulic Engineering for a resilient and sustainable future

Vision:

We believe Hydraulic Engineering is at the core of a safe, resilient, and sustainable civil system, where people and the environment thrive. Our focus is on enhancing the design and performance of hydraulic systems, improving infrastructure, strengthening climate adaptation and hazard resilience, and accelerating the transition to clean energy.

Mission:

We fulfil this vision through world-leading education, research and valorisation in the field of hydraulic engineering. Through our unique education we empower the next generation of hydraulic engineers to bring new knowledge and innovation to the field. Our cutting-edge research is grounded in a deep understanding of hydraulic processes and system dynamics, synergising theory, modelling, fieldwork and laboratory experiments. We achieve real world impact and address global challenges by actively collaborating with governments, industry, and academic partners to deliver practical and sustainable engineering solutions for society.

We embrace the TU Delft values by fostering an academic leadership and working environment that promotes diversity, inclusion, equity, integrity and courage.

Give feedback on the document

Summary session 1+2

You can find the whole overview here

Achievement	Activities towards achievement	Measures of Success
1. We conduct scientifically excellent and societally relevant research that has impact	Publish papers of higher “value” and that push the state of art	Number of high-impact (high citation) peer reviewed publications and patents
	Use models as knowledge integrators	Position in Global Rankings
	Build academic collaborations that promote excellent science and teamwork.	Success in competitive grants (e.g. NWO, ERC)
	Embed societal impact into our strategic agenda/s	Research uptake in policy/engineering practice
	Be aware and respond to national and international agendas/roadmaps	Number of active PhDs/postdocs per section (as indicator of research strength and continuity)
	Conduct research that has the potential for application/is applied	Staff involved in International code committees/advisory groups/professional organisations
	Ensure the right type (skills, seniority) and number of human resources are employed on projects/research	No. of downloads of tools and datasets
	Actively monitor impact e.g. How our research is used by others and who uses it	Amount of media attention
	Build strong collaborations with NGO’s, industry and government organisations	Research awards and other recognition (e.g Key Notes, invited lectures, honorary roles)

Activity 1:HE research/education lines

Activity 1- HE research/education lines

Assignment:

Imagine the ideal TU Delft Hydraulic Engineering department from three different viewpoints.

For each perspective, identify which research/education lines or topics you would expect a Hydraulic Engineering department to cover.

Perspectives:

- Prospective students.
- Dutch government/ Industry partners.
- Global academic community.

Instructions

- Discuss each perspective one by one.
- For each perspective, identify research/education lines/topics you would expect to find in an HE department.
- Capture your ideas the flip-over.
- Be as concrete as possible (e.g., “coastal resilience,” “sediment dynamics,” “AI for hydraulic modelling”).

Timing

- 20 minutes – group activity
- 20 minutes – plenary feedback

BREAK TIME

Activity 2: Designing the HE department of the future

Activity 2A - Designing the HE Department of the future

Assignment – Part A:

Using the outcomes of round 1, design your ideal version of the Hydraulic Engineering department.

Propose an organisational structure that supports:

- The research/education lines you identified
- The HE vision and Mision
- the 7 HE achievements

Instructions:

- Use the organisation format to define your proposed sections.
- Create section names and list the research/education topics that belong to each section.

Timing:

- 25 minutes group work

Activity 2 B- Designing the HE Department of the future

Assignment:

Justify your HE Department Design

Instructions:

- Evaluate your design by filling in the SWOT analysis

Timing:

- 20 minutes group work
- 2-minute pitch per group

Next steps

- We'll process your input from both activities and share a summary with the department.
- **Next session – 29 January**
  - The board will prepare a proposal based on the input of session 1,2 and 3.
- This proposal will be presented for discussion and refinement

End of the session:

Thank you for your input!